

6 Enterprise risk management at Shell

Shell plc is a global energy company with operations covering upstream exploration and production, integrated gas, chemicals, trading, power generation, and retail activities. Its business model exposes it to a wide range of uncertainties, including commodity price volatility, geopolitical risk, operational and safety hazards, regulatory change, and long-term transition risks related to climate and energy policy. The scale of its operations, combined with capital intensity and global reach, makes risk management a central component of corporate governance.

Shell's current approach to enterprise risk management has been shaped by historical crises, including major operational incidents and periods of financial stress. These events revealed weaknesses in governance, escalation, and oversight, and led to a deliberate strengthening of group-level risk management. As a result, Shell has positioned ERM not as a purely technical or compliance-driven activity, but as an integral part of decision-making and strategic oversight.

At governance level, the Board of Directors retains ultimate responsibility for risk management and internal control. Specific oversight responsibilities are delegated to board committees. The Audit and Risk Committee reviews the effectiveness of Shell's risk management and internal control systems, while the Safety, Environment and Sustainability Committee focuses on risks related to safety performance, environmental impact, and the energy transition. This structure reflects an explicit recognition that non-financial risks can be as material as financial risks.

Within management, Shell operates a centralized enterprise risk management framework. Risk ownership is assigned to senior executives and business leaders, who remain accountable for the risks arising from their activities. The central risk function does not own risks; instead, it provides common methodologies, coordination across the group, and independent challenge. This separation between ownership and oversight is intended to ensure that risk remains embedded in line management rather than delegated to specialists.

Shell identifies a limited set of *principal risks* that are disclosed annually in its reporting. These principal risks represent those that could materially affect the company's strategy, performance, financial position, or reputation. They include, among others, commodity price risk, geopolitical and regulatory risk, safety and

operational incidents, climate transition risk, cyber risk, and liquidity and funding risk. The list is not static and evolves as the external environment and strategic priorities change.

Risk assessment combines quantitative analysis with qualitative judgment and scenario-based thinking. For market-related risks, particularly commodity price exposure, Shell relies on quantitative metrics, stress testing, and risk limits within its trading and treasury activities. For longer-term and structural uncertainties, such as the pace of the energy transition or major geopolitical shifts, Shell uses scenario analysis rather than point forecasts. These scenarios are designed to test strategic resilience rather than predict specific outcomes.

Risk responses are tailored to the nature of each principal risk. Operational risks are addressed through safety management systems, engineering standards, and process controls applied across the group. Financial risks, including commodity price, foreign exchange, and interest rate risks, are managed through a combination of natural hedging, portfolio diversification, and the use of derivative instruments. Derivatives are widely used in trading and treasury activities, but their use is governed by formal policies that define eligible instruments, limits, approval processes, and escalation thresholds. Hedging is intended to support the underlying business and protect cash flows rather than to eliminate all exposure.

Liquidity risk management is treated as a core enterprise risk. Shell maintains substantial liquidity buffers and diversified funding sources to ensure resilience under stress. The firm explicitly considers the interaction between derivative hedging and liquidity, including the potential for margin calls during periods of market volatility. This reflects an awareness that hedging strategies can create short-term funding stress even when they are economically sound over the long term.

Enterprise risk management is embedded in Shell's strategic planning and capital allocation processes. Major investment decisions are evaluated not only on expected returns but also on their robustness under adverse scenarios and consistency with risk appetite and balance-sheet capacity. Long-term risks, particularly those related to climate and energy transition, are treated as strategic uncertainties that cannot be hedged through financial instruments. Instead, they are managed through portfolio diversification, staged investment, and strategic flexibility.

Risk reporting is designed to support escalation and decision-making rather

than passive monitoring. Reports to senior management and the board focus on key risk drivers, changes in the risk profile, and proximity to defined risk tolerances. When indicators deteriorate or approach limits, management is expected to respond by adjusting plans, revising exposures, or explicitly accepting the risk at the appropriate governance level.

Overall, Shell's enterprise risk management system reflects a principles-based and integrated approach. Risk management is embedded in governance, strategy, and capital allocation rather than operating as a separate technical layer. At the same time, the effectiveness of the system depends heavily on the quality of assumptions, the willingness of leaders to engage with challenge, and the organization's ability to adapt when familiar risk patterns break down. The case therefore illustrates both how ERM is implemented in practice by a large multinational firm and why even sophisticated ERM cannot eliminate the possibility of failure.

Discussion questions

Answer the following questions using evidence from the case above. Answers should focus on reasoning and diagnosis rather than description.

ERM framework and philosophy

1. How would you characterize Shell's ERM framework? Is it primarily rules-based or principles-based? Compliance-oriented or decision-oriented? Which elements of the case support your view?
2. Shell focuses on a limited number of *principal risks*. What does this reveal about its ERM philosophy, and what are the potential trade-offs of this approach?
3. Why does Shell rely heavily on scenario analysis rather than point forecasts for long-term risks? What does this imply about the limits of quantitative risk measurement?

ERM architecture and governance

4. Identify the key components of Shell's ERM architecture. How are risk ownership and risk oversight allocated across the organization?
5. Why is the separation between business risk ownership and the central ERM function emphasized? Under what circumstances could this separation weaken risk management?
6. How might the existence of multiple board-level committees complicate coordination and escalation of risks?

Risk integration and instruments

7. Identify several risk categories addressed in Shell's ERM system. How does the case suggest these risks are considered in an integrated way?
8. What role do derivative instruments play in Shell's risk management? Why are formal constraints on their use essential from an ERM perspective?
9. Why does Shell explicitly consider the liquidity implications of derivative hedging? What type of failure is this designed to prevent?

Strategy, limits, and failure modes

10. How is ERM embedded in Shell's strategic planning and capital allocation processes? Give concrete examples from the case.
11. Which aspects of Shell's ERM design appear most dependent on managerial judgment rather than formal rules? Why does this matter?
12. Describe a plausible scenario in which Shell's ERM system could fail despite being well designed. What would trigger such a failure, and which mechanisms would likely break down first?
13. Based on the case, explain why effective ERM reduces vulnerability but does not eliminate the possibility of failure.